

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');

```

```

fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off200GHmaxFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');

```

```

*****

```

```

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei800_off200GHmaxFF

```

```

fprintf('*****\n');

```

```

*****

```

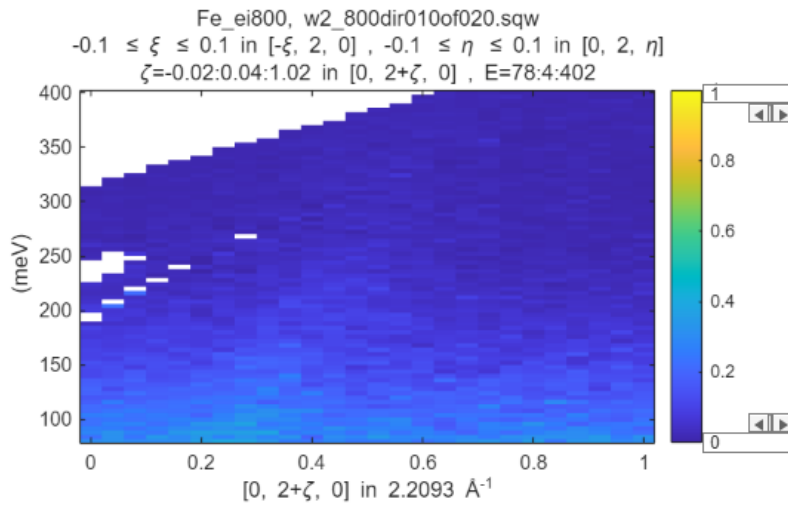
```

cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

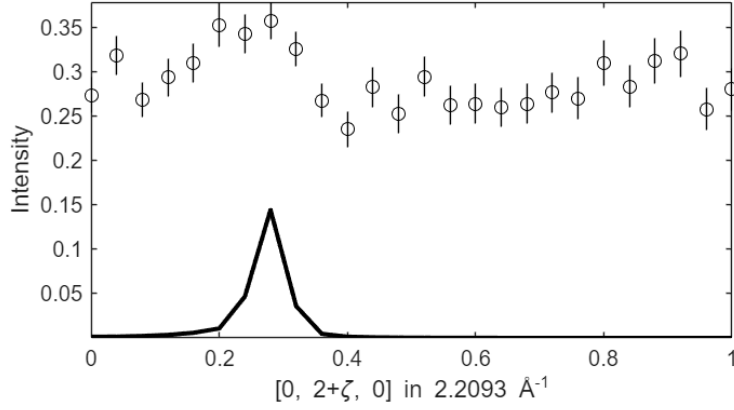
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFcutsFits.(selectCuts) = cuts_list;

GH_FFcutsFits = fit_multicutsFF_along_direction(...
    GH_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

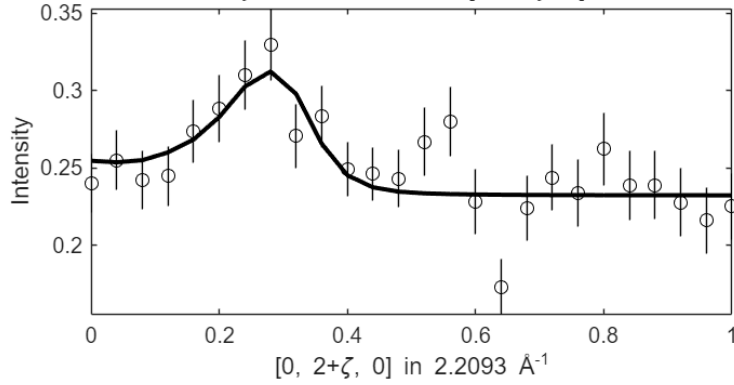
```
*****
**** En = 83±5 going to 393
*****
```

Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $78 \leq E \leq 8$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

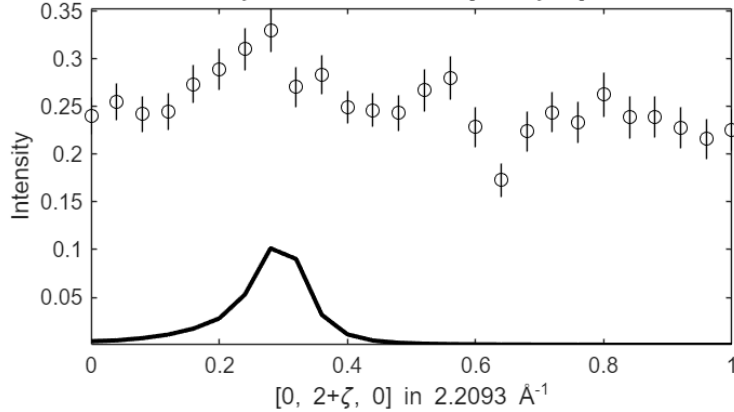


 **** En = 93±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.74 \pm 0.3$; $J_0= 30 \pm 3.1$; $J_1= 0 \pm 0$; $\gamma=26.6148 \pm 21.754$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $88 \leq E \leq 9$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

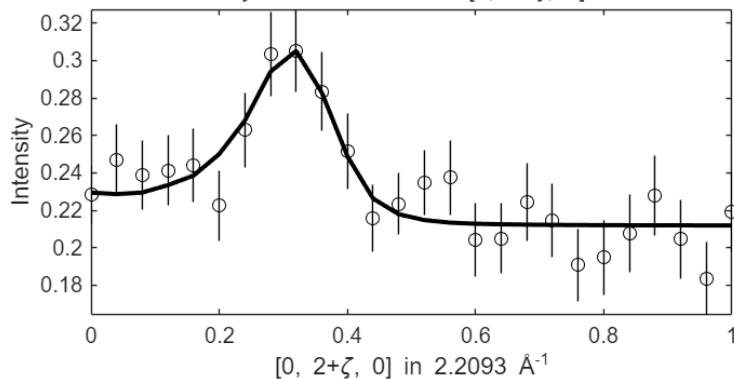


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $88 \leq E \leq 9$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

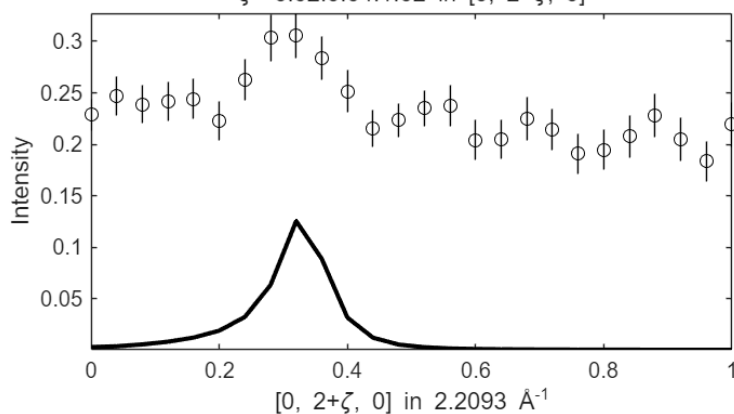


 **** En = 103±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.81\pm0.072$; $J_0= 27\pm 1.2$; $J_1= 0\pm 0$; $\gamma=23.0128\pm1.86228$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $98 \leq E \leq 113$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

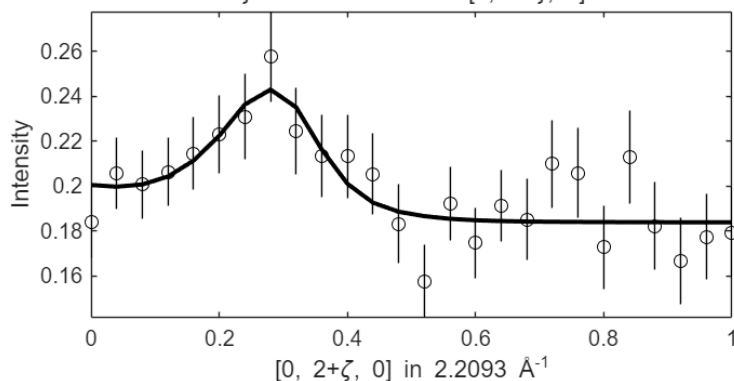


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $98 \leq E \leq 113$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

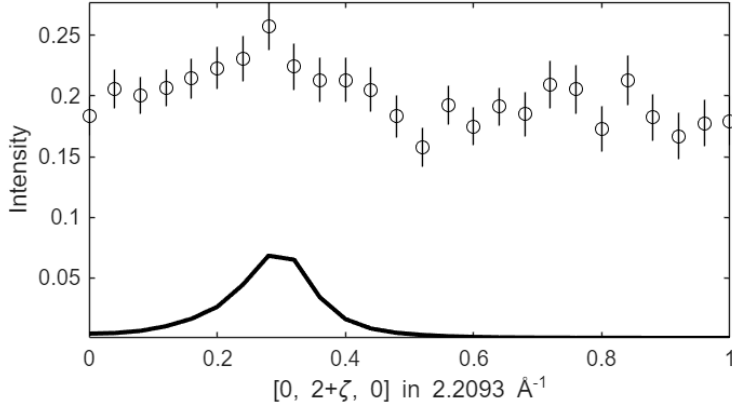


 **** En = 113±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.78\pm0.094$; $J_0= 34\pm 2.7$; $J_1= 0\pm 0$; $\gamma=43.1164\pm5.00476$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $108 \leq E \leq 113$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

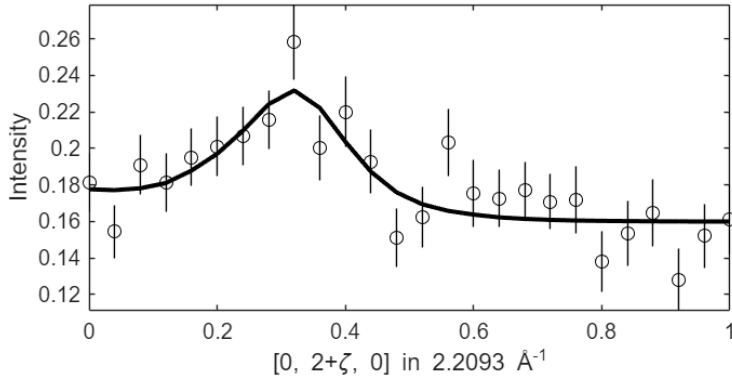


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $108 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

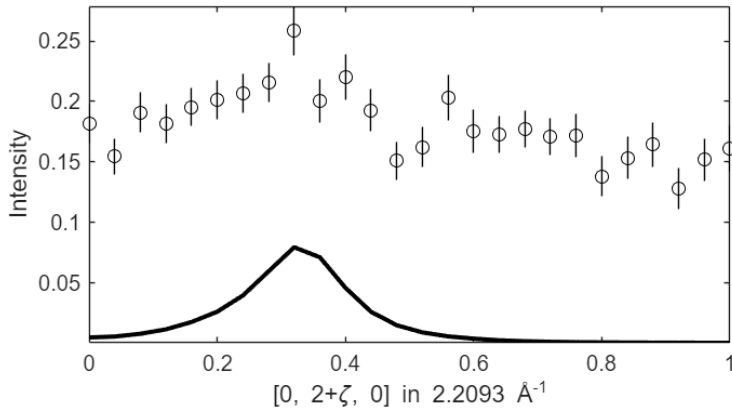


 **** En = 123±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.3 \pm 0.54$; $J_0 = 31 \pm 3.5$; $J_1 = 0 \pm 0$; $\gamma = 60.8248 \pm 36.9287$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $118 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

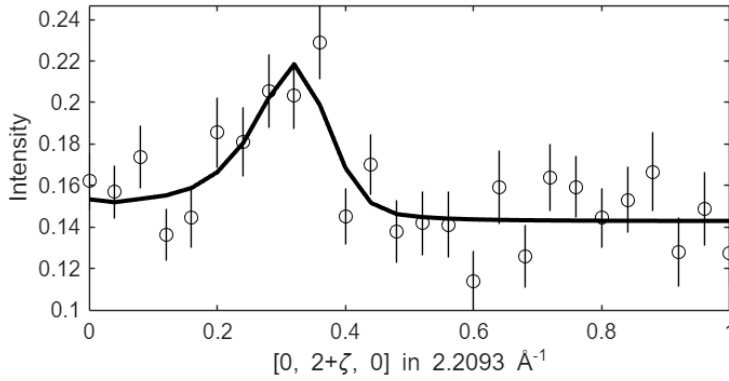


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $118 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

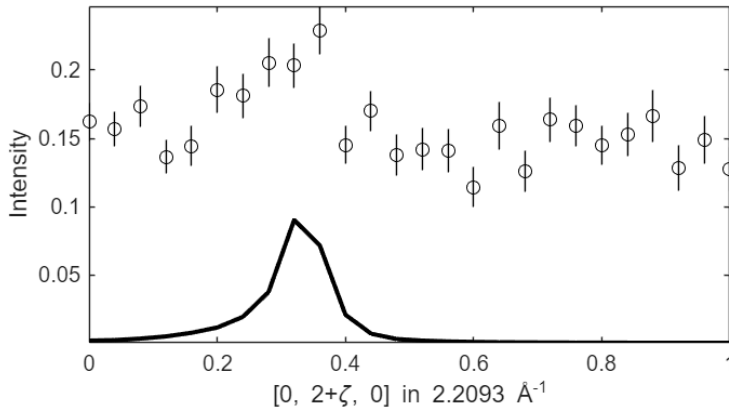


 **** En = 133±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.77 \pm 0.1$; $J_0= 34 \pm 2.3$; $J_1= 0 \pm 0$; $\gamma=24.6578 \pm 3.14172$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $128 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

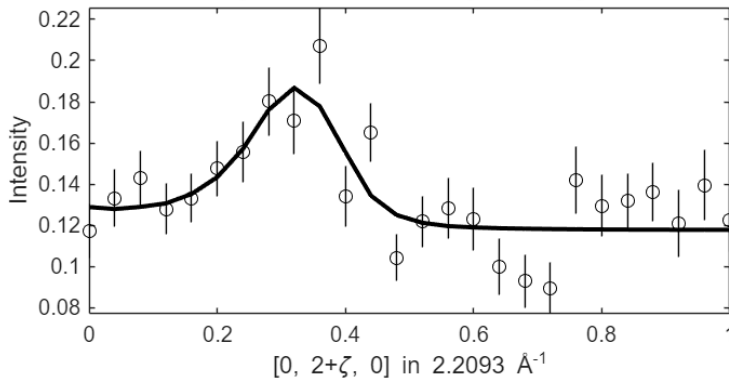


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $128 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

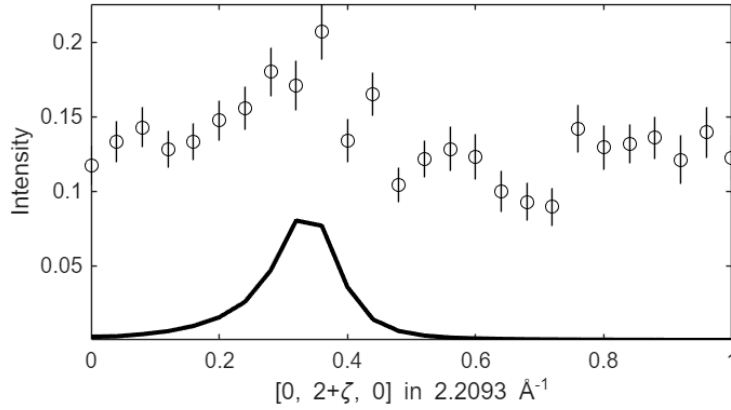


 **** En = 143±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.91 \pm 0.13$; $J_0= 35 \pm 2.6$; $J_1= 0 \pm 0$; $\gamma=37.7467 \pm 4.75361$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $138 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

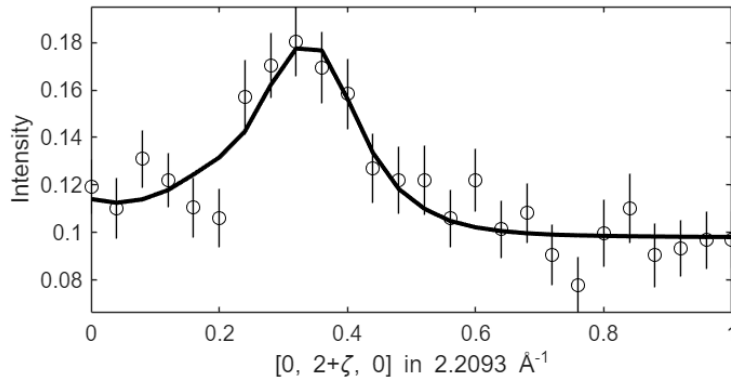


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $138 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

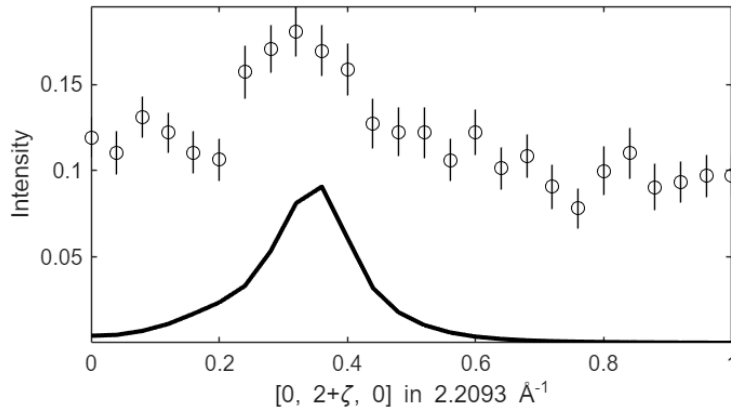


**** En = 153±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.6 \pm 0.13$; $J_0 = 35 \pm 1.7$; $J_1 = 0 \pm 0$; $\gamma = 60.9923 \pm 3.52877$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $148 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



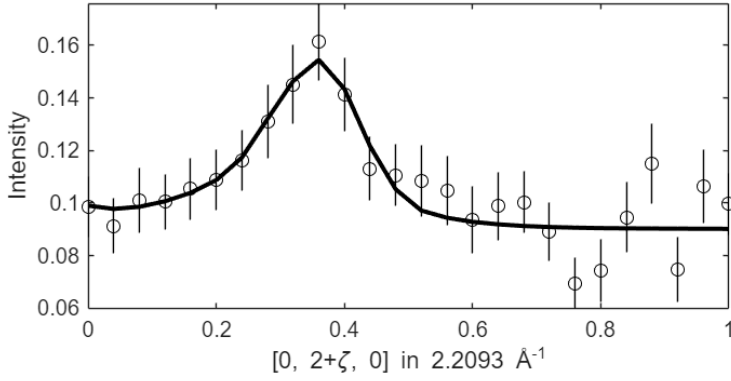
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $148 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



**** En = 163±5 going to 393

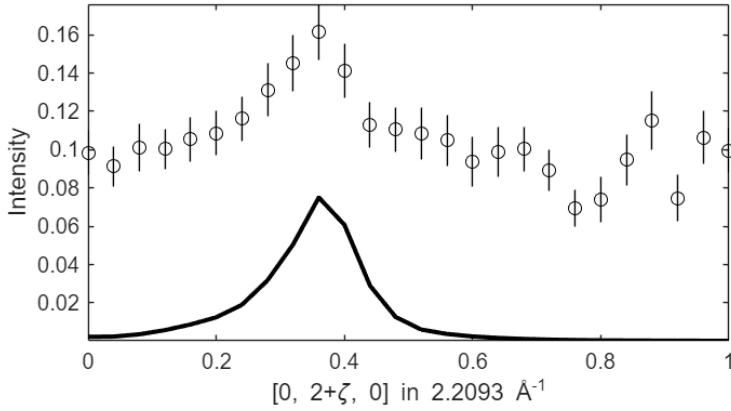
Fe_ei800, w2_800dir010of020.sqw

S= 1.1 ± 0.25 ; J0= 34 ± 1.7 ; J1= 0 ± 0 ; gamma= 47.0004 ± 17.685
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $158 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

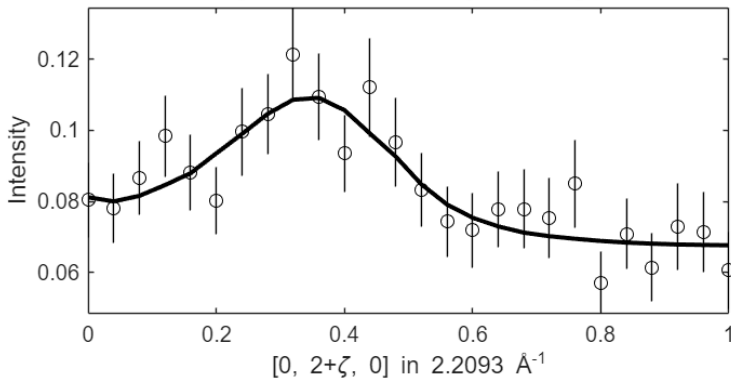
S= 1.1 ± 0.25 ; J0= 34 ± 1.7 ; J1= 0 ± 0 ; gamma= 47.0004 ± 17.685
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $158 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



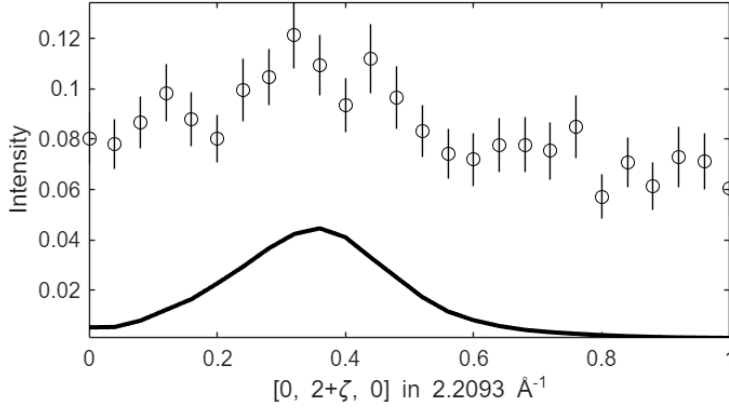
**** En = 173 ± 5 going to 393

Fe_ei800, w2_800dir010of020.sqw

S= 2.5 ± 2 ; J0= 48 ± 19 ; J1= 0 ± 0 ; gamma= 241.274 ± 267.38
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $168 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

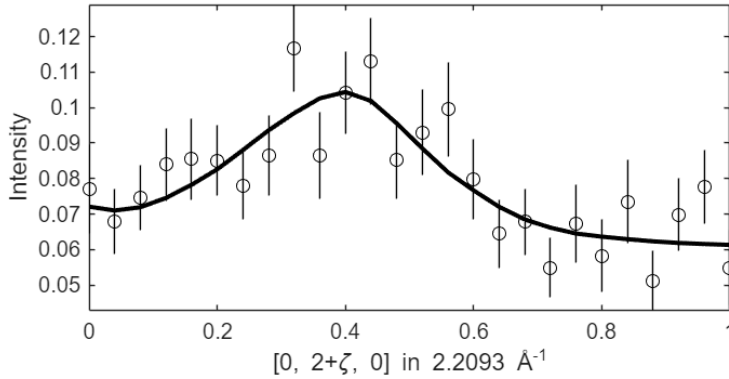


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $168 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

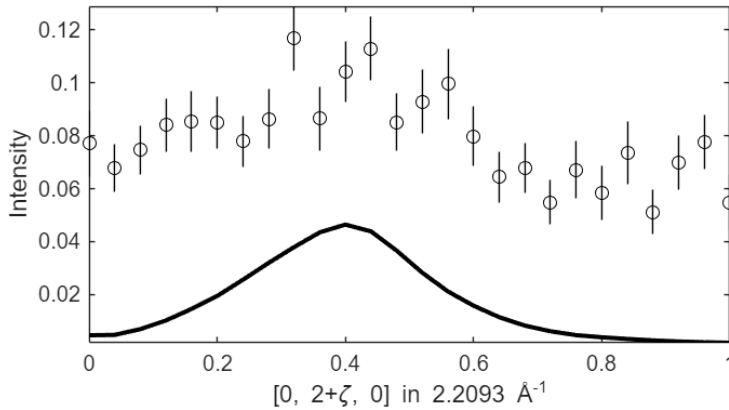


**** En = 183±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 2.6 \pm 2.1$; $J_0 = 39 \pm 13$; $J_1 = 0 \pm 0$; $\gamma = 208.733 \pm 223.593$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $178 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

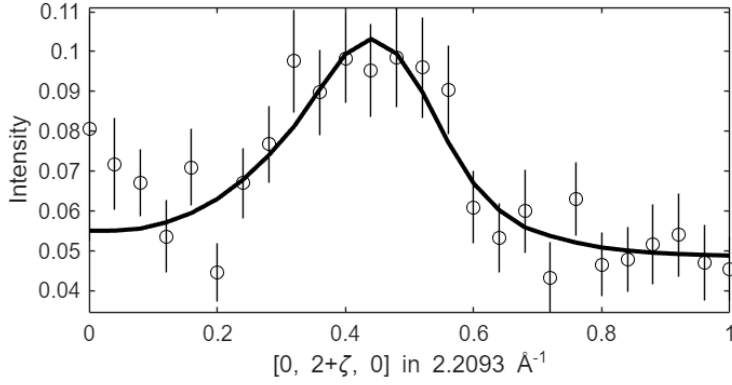


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $178 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

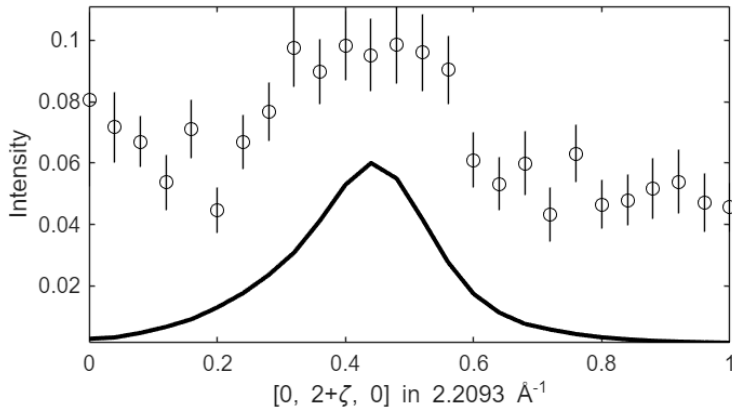


**** En = 193±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.8 \pm 0.18$; $J_0 = 30 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 89.94 \pm 5.27257$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $188 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

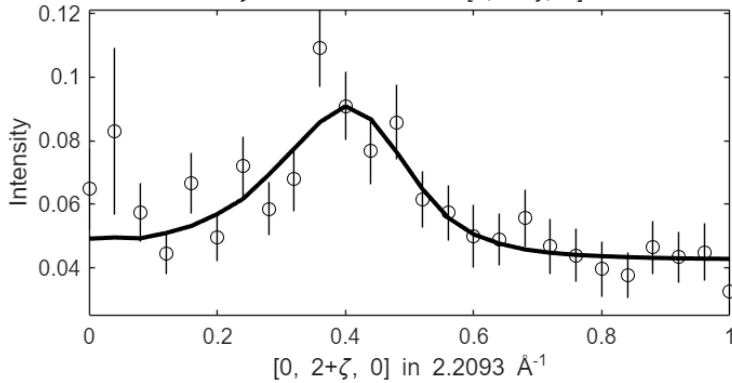


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $188 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

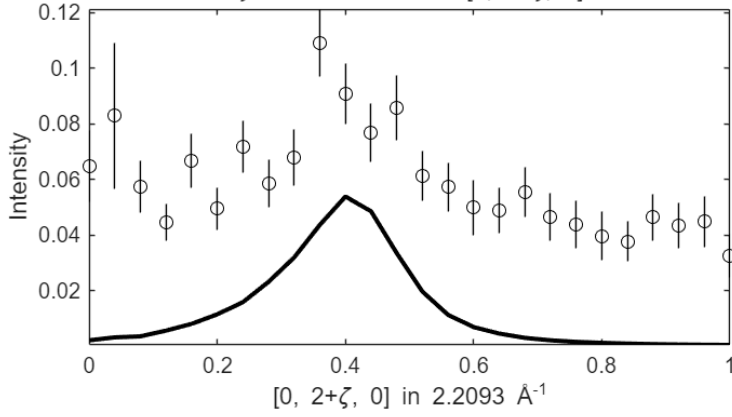


 **** En = 203±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.4 \pm 0.14$; $J_0 = 36 \pm 2.1$; $J_1 = 0 \pm 0$; $\gamma = 83.4654 \pm 6.79069$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $198 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

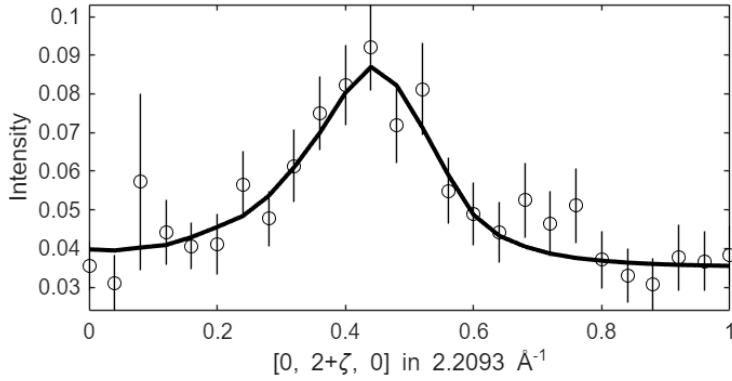


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $198 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

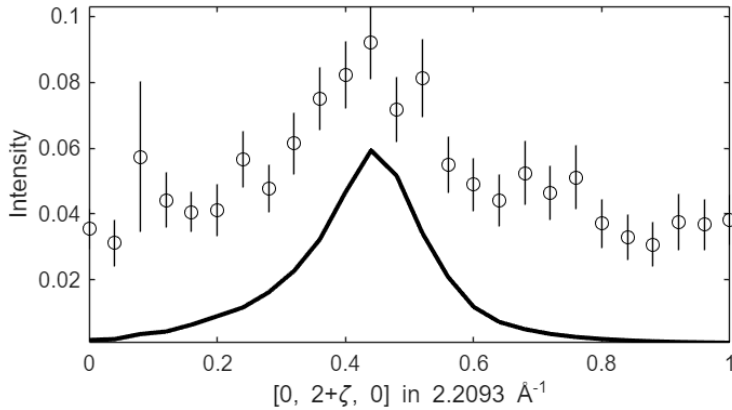


 **** En = 213±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.5 \pm 0.1$; $J_0 = 32 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 71.28 \pm 3.59152$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $208 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



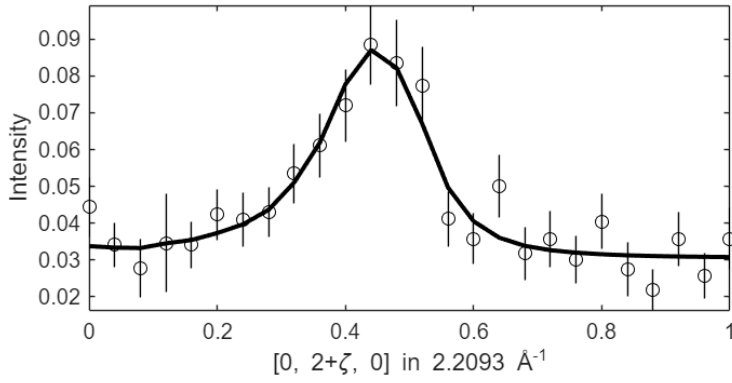
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $208 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 223±5 going to 393

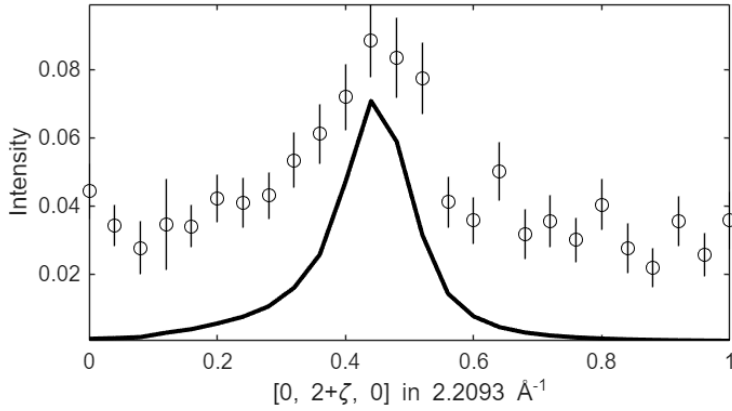
Fe_ei800, w2_800dir010of020.sqw

$S = 1.2 \pm 0.087$; $J_0 = 33 \pm 1.2$; $J_1 = 0 \pm 0$; $\gamma = 46.9788 \pm 2.76765$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $218 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

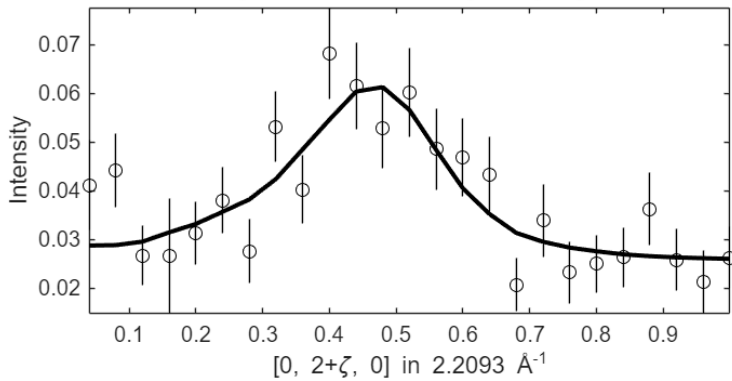
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $218 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



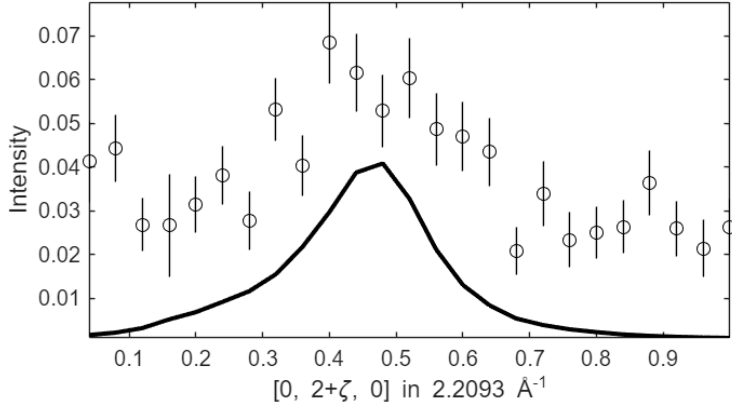
 **** En = 233±5 going to 393

Fe_ei800, w2_800dir010of020.sqw

$S = 1.3 \pm 0.16$; $J_0 = 33 \pm 2.1$; $J_1 = 0 \pm 0$; $\gamma = 85.9725 \pm 7.32074$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $228 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

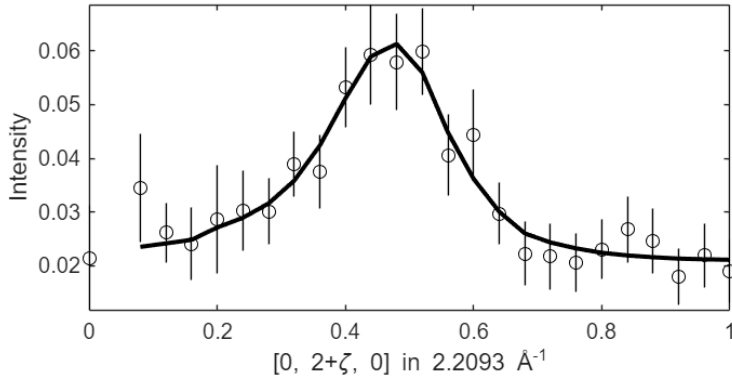


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $228 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

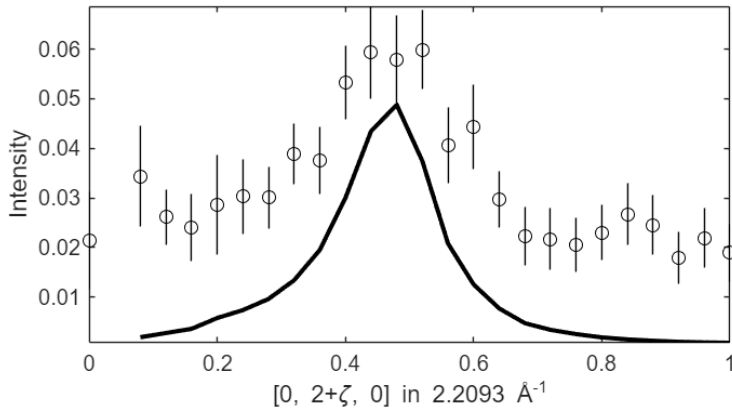


 **** En = 243±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 S= 1.2±0.076; J0= 33±0.82; J1= 0± 0; gamma=68.7177±2.6196
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $238 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



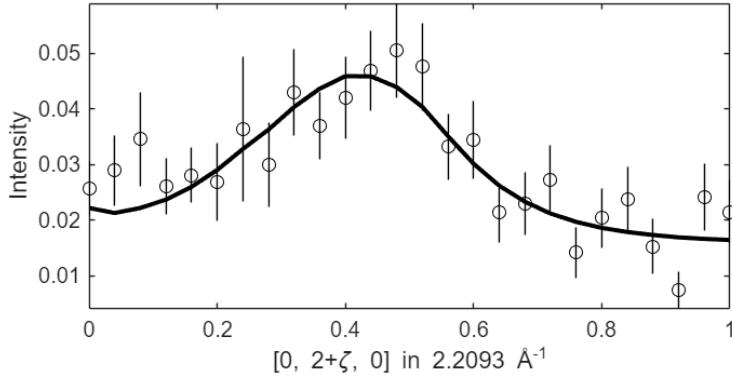
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $238 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 253±5 going to 393

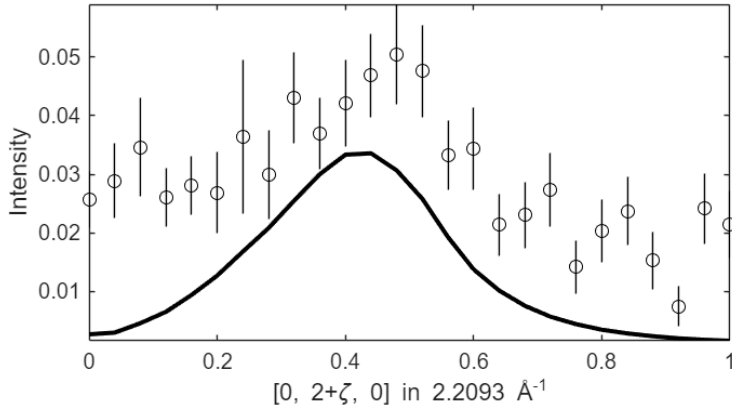
Fe_ei800, w2_800dir010of020.sqw

S= 2.5 ± 1.8 ; J0= 47 ± 12 ; J1= 0 ± 0 ; gamma= 248.427 ± 229.457
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $248 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

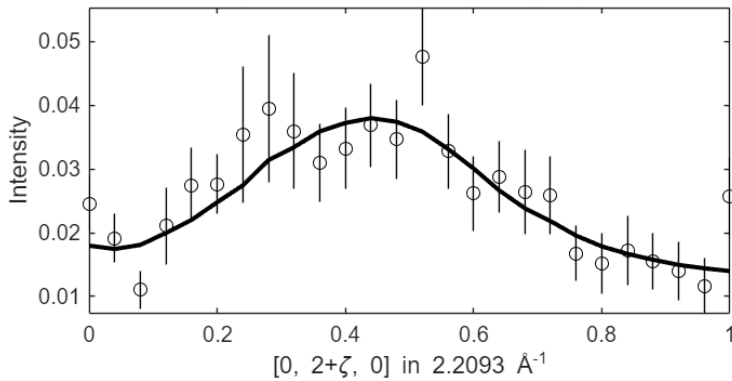
S= 2.5 ± 1.8 ; J0= 47 ± 12 ; J1= 0 ± 0 ; gamma= 248.427 ± 229.457
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $248 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



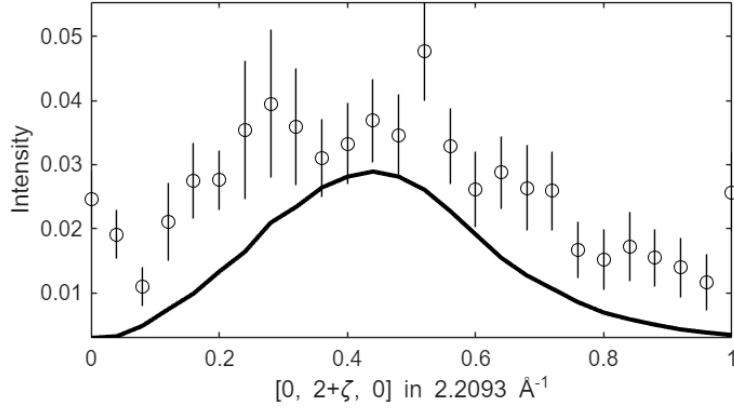
**** En = 263 ± 5 going to 393

Fe_ei800, w2_800dir010of020.sqw

S= 6 ± 19 ; J0= $72 \pm 1.7e+02$; J1= 0 ± 0 ; gamma= 1012.81 ± 5355.96
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $258 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

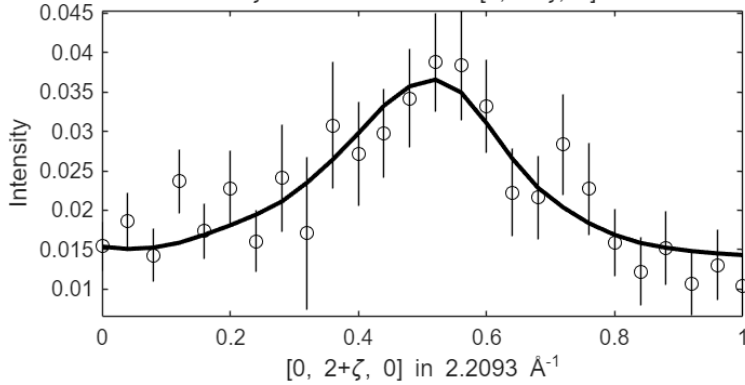


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $258 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

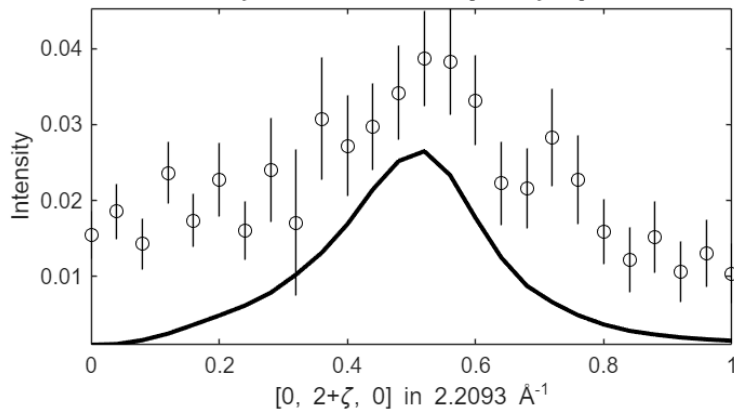


 **** En = 273±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 1.2 \pm 0.13$; $J_0 = 34 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 118.739 \pm 7.0242$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $268 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



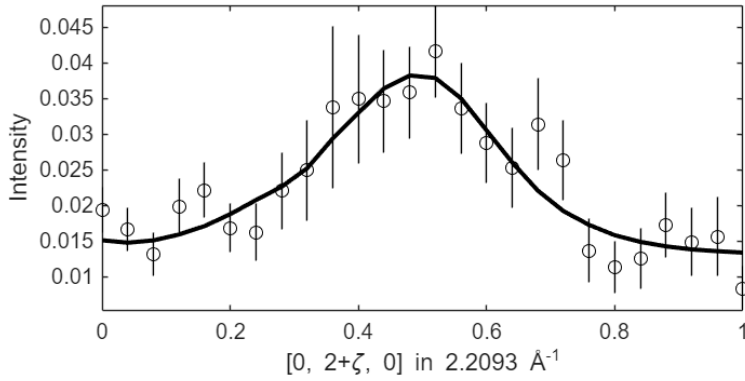
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $268 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 283±5 going to 393

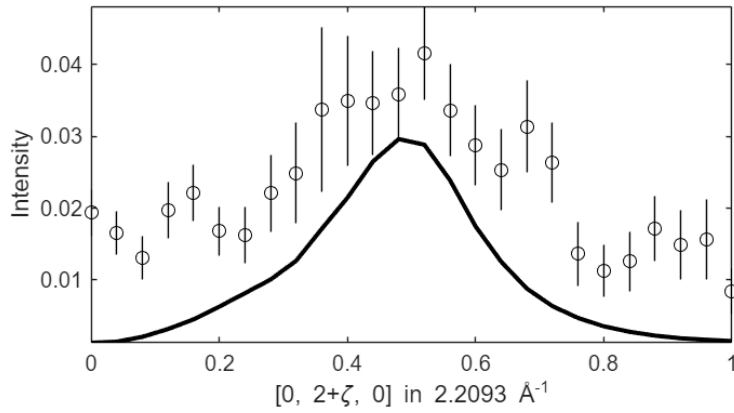
Fe_ei800, w2_800dir010of020.sqw

S= 1.5 ± 0.51 ; J0= 37 ± 2.4 ; J1= 0 ± 0 ; gamma= 134.106 ± 56.2674
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $278 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

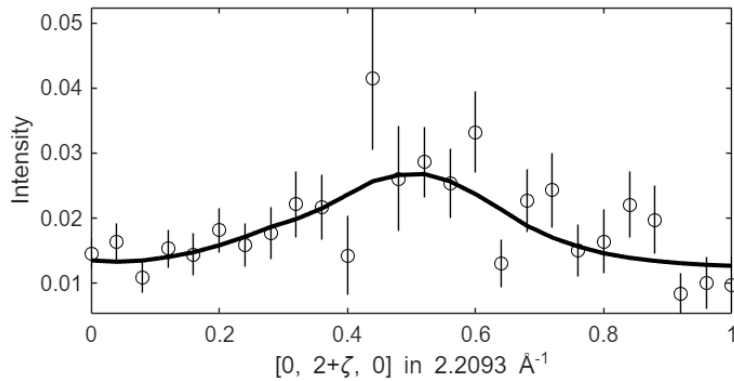
S= 1.5 ± 0.51 ; J0= 37 ± 2.4 ; J1= 0 ± 0 ; gamma= 134.106 ± 56.2674
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $278 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



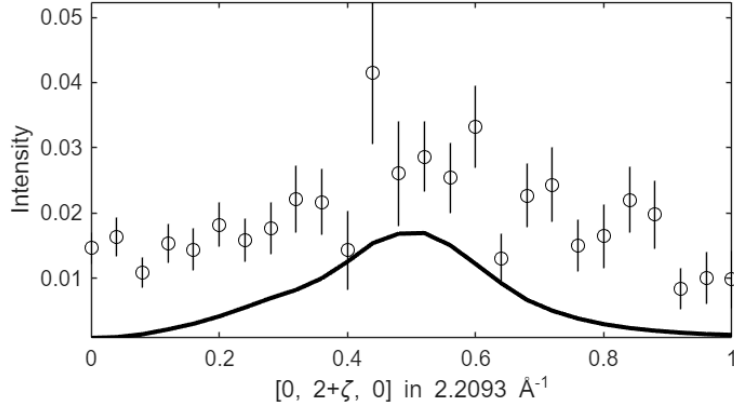
**** En = 293 ± 5 going to 393

Fe_ei800, w2_800dir010of020.sqw

S= 1.1 ± 0.19 ; J0= 38 ± 3.2 ; J1= 0 ± 0 ; gamma= 167.502 ± 16.8558
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $288 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

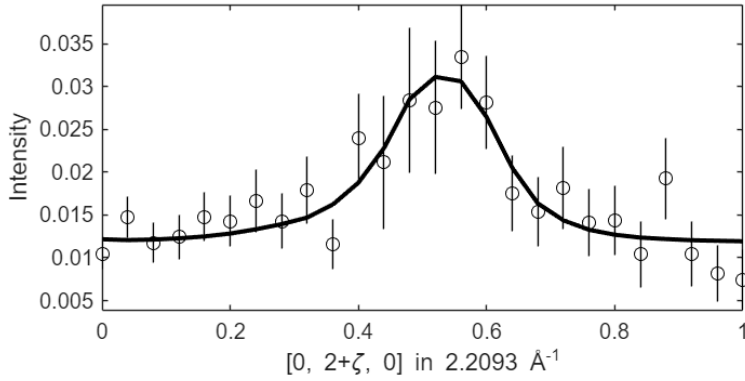


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $288 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

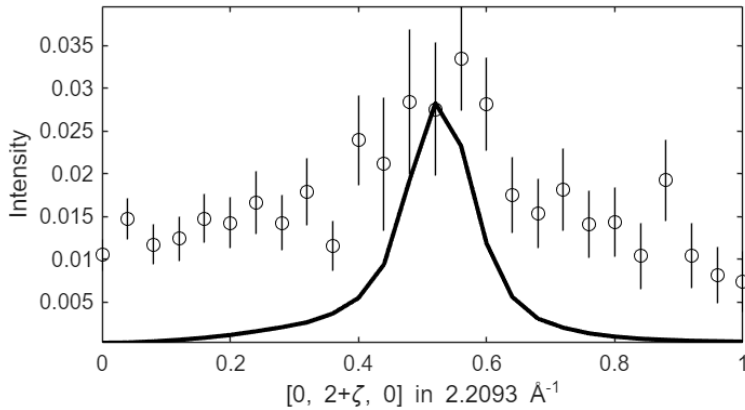


 **** En = 303±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S = 0.59 \pm 0.14$; $J_0 = 35 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 48.3949 \pm 22.342$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $298 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



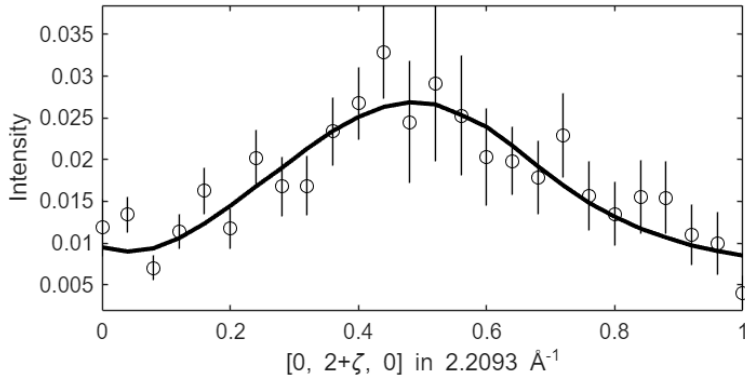
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $298 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 313±5 going to 393

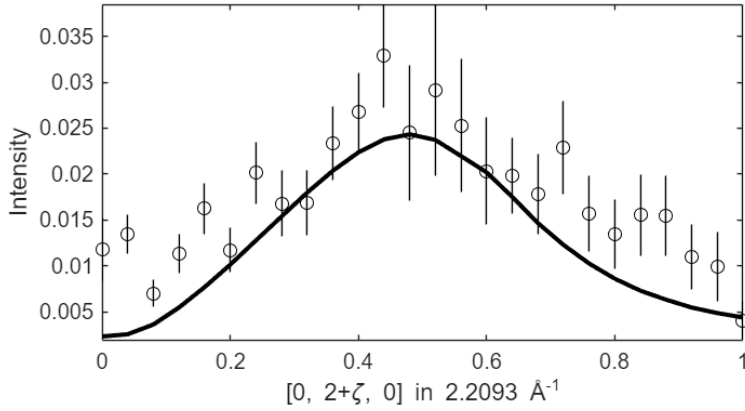
Fe_ei800, w2_800dir010of020.sqw

S= 4.9± 8; J0= 60± 60; J1= 0± 0; gamma=740.1±1858.44
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $308 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

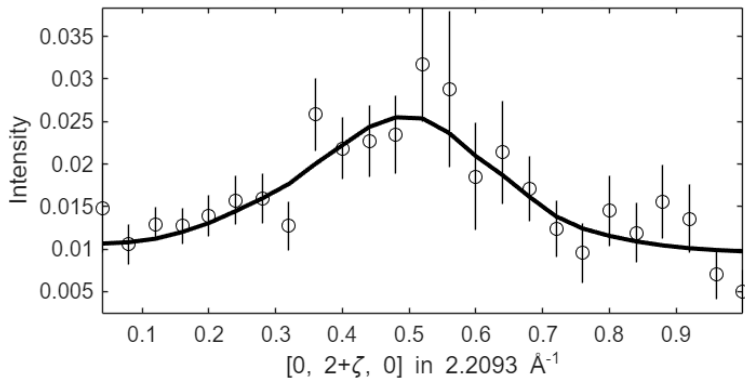
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $308 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



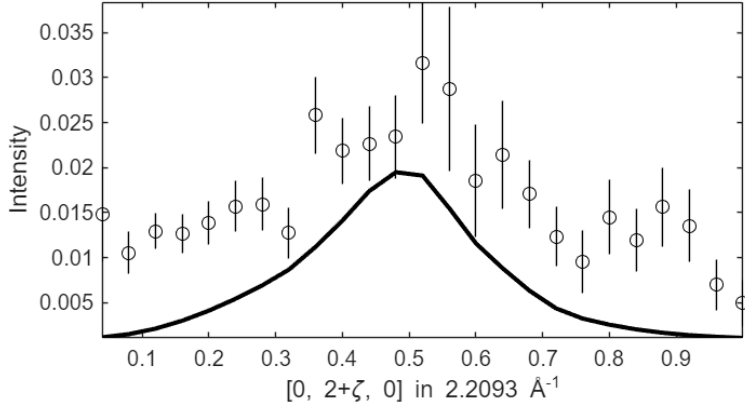
**** En = 323±5 going to 393

Fe_ei800, w2_800dir010of020.sqw

S= 1.2±0.16; J0= 42± 2.4; J1= 0± 0; gamma=156.686±10.3227
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $318 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

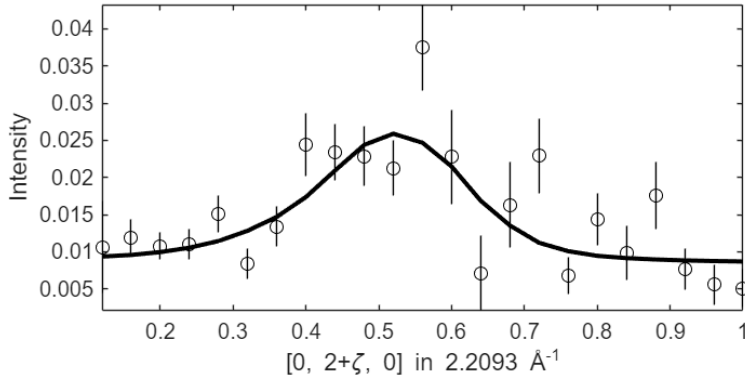


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $318 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

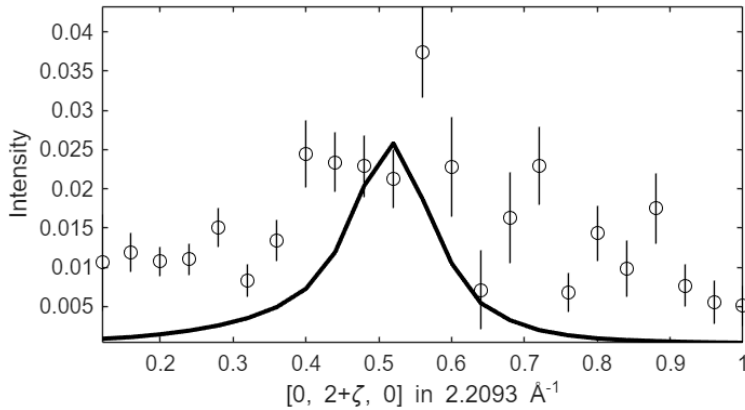


 **** En = 333±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.68 \pm 0.12$; $J_0 = 40 \pm 3.1$; $J_1 = 0 \pm 0$; $\gamma = 63.5475 \pm 7.82799$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $328 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

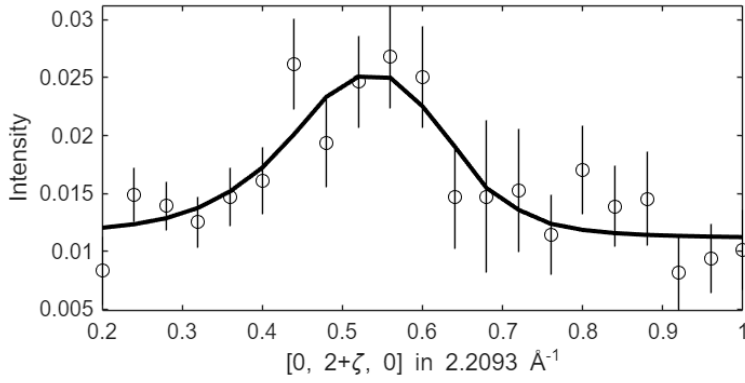


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $328 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

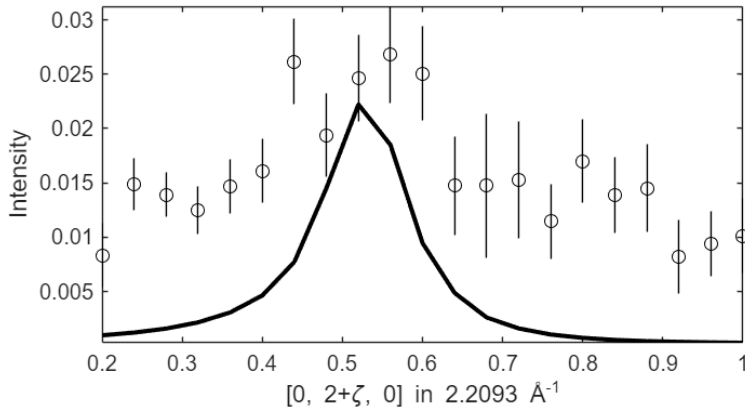


 **** En = 343±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.52\pm0.062$; $J_0= 39\pm 2.1$; $J_1= 0\pm 0$; $\gamma=54.2066\pm5.05431$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $338 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

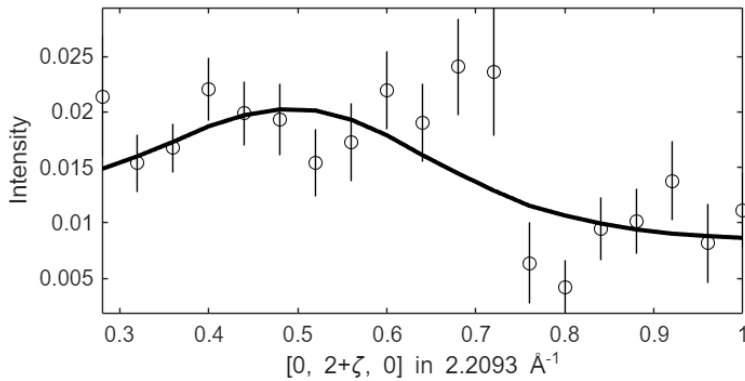


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $338 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

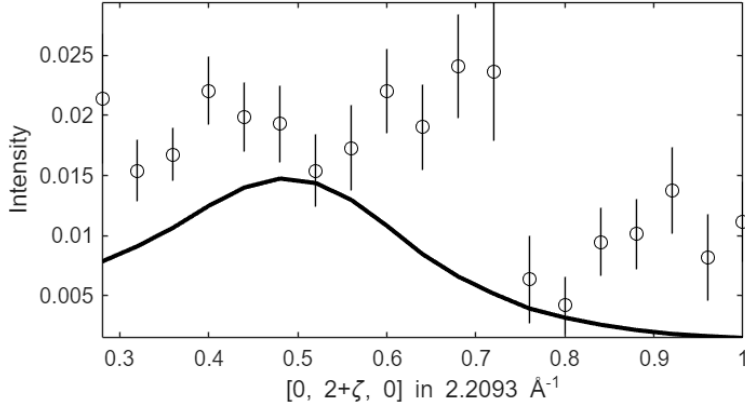


 **** En = 353±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S= 1.5\pm0.58$; $J_0= 50\pm 5.9$; $J_1= 0\pm 0$; $\gamma=293.668\pm37.3416$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $348 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

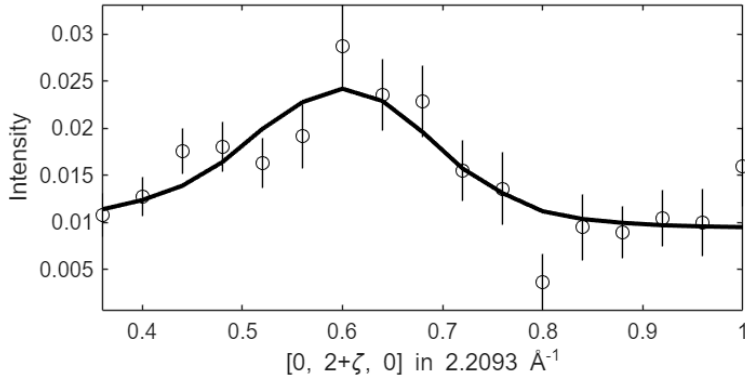


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $348 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

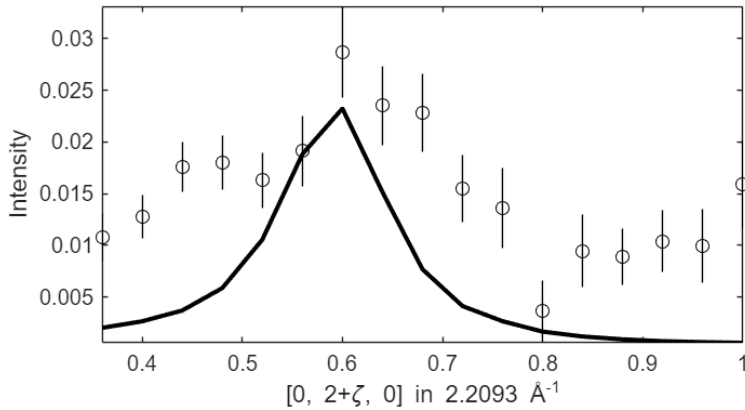


**** En = 363±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.56 \pm 0.092$; $J_0= 35 \pm 1.3$; $J_1= 0 \pm 0$; $\gamma=49.0196 \pm 4.67707$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $358 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

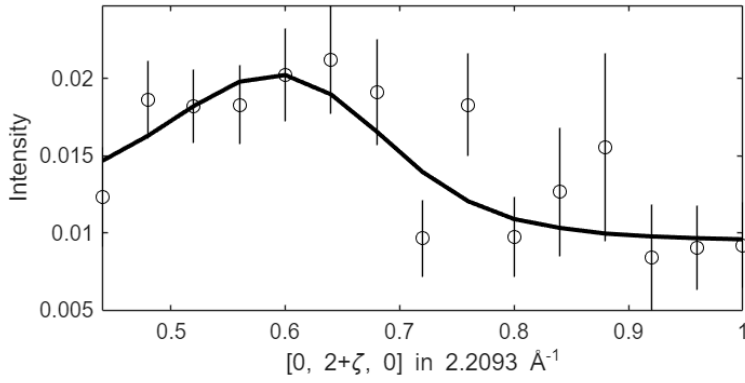


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $358 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

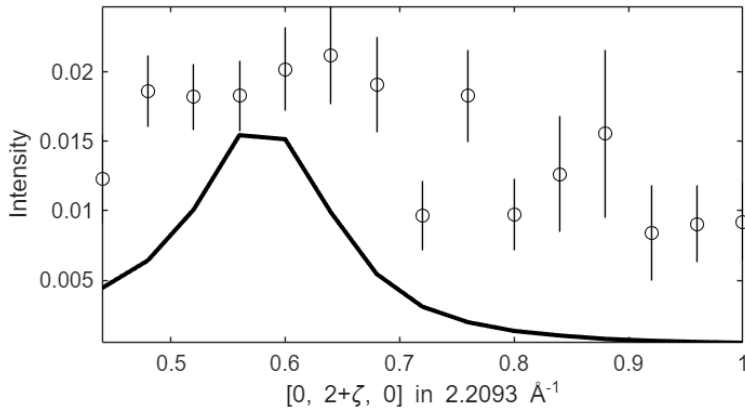


**** En = 373±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.48\pm0.092$; $J_0= 38\pm 2.1$; $J_1= 0\pm 0$; $\gamma=62.7568\pm5.3261$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $368 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

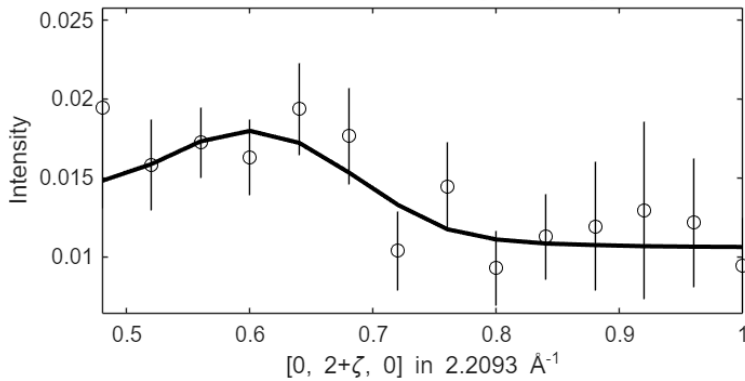


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $368 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

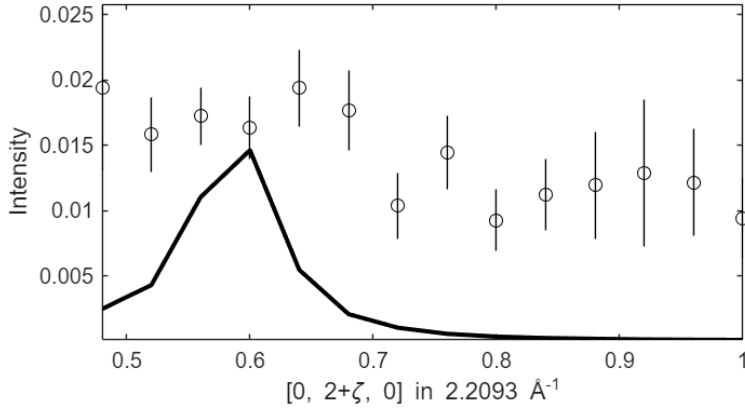


 **** En = 383 ± 5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 $S=0.24\pm0.043$; $J_0= 38\pm 2.3$; $J_1= 0\pm 0$; $\gamma=29.704\pm4.22956$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $378 \leq E \leq 3$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

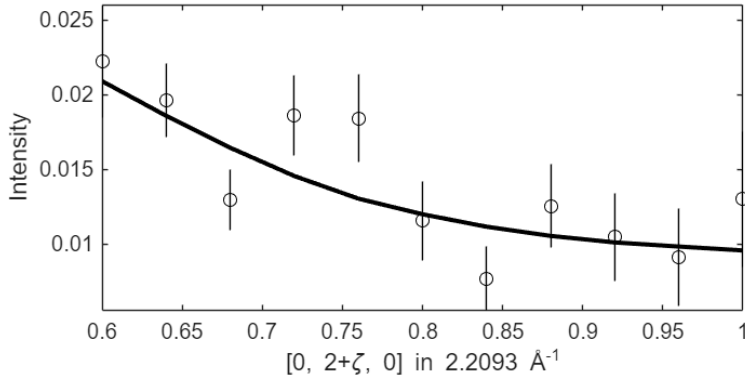


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $378 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

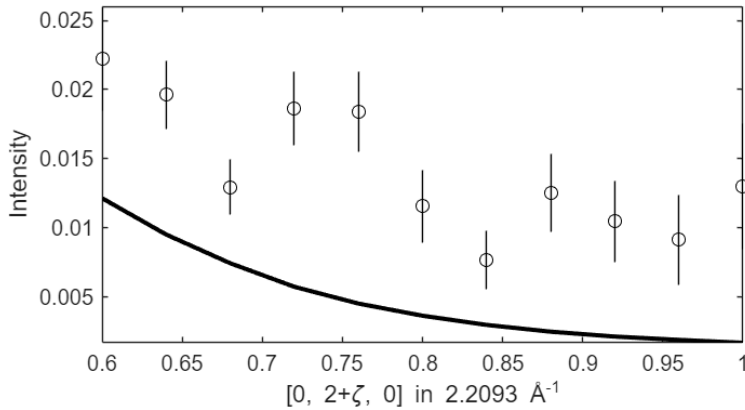


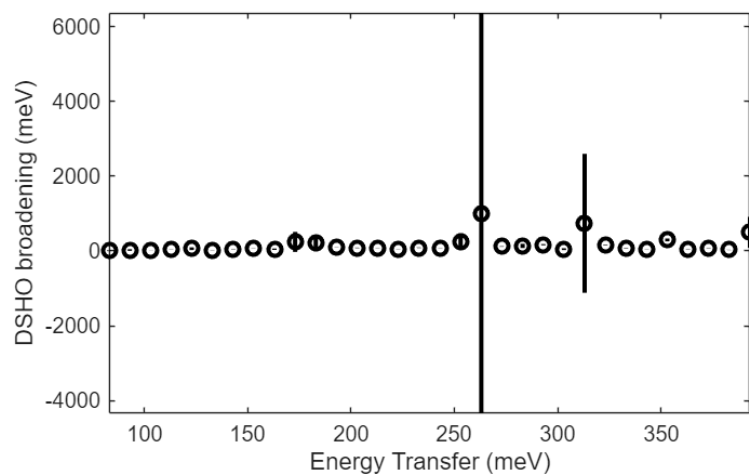
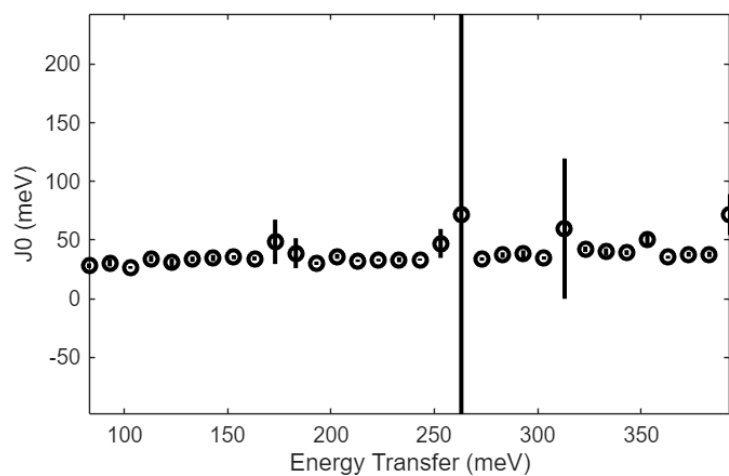
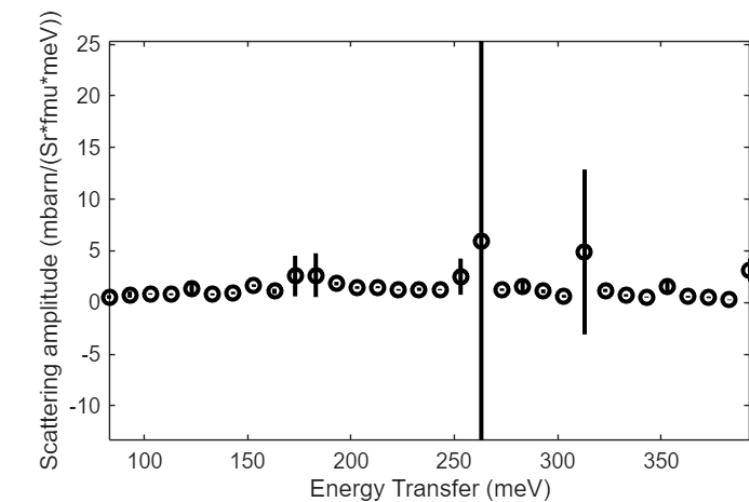
 **** En = 393±5 going to 393

Fe_ei800, w2_800dir010of020.sqw
 S= 3.1± 1.1; J0= 72± 18; J1= 0± 0; gamma=500.273±413.82
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $388 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $388 \leq E \leq 3$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$





```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

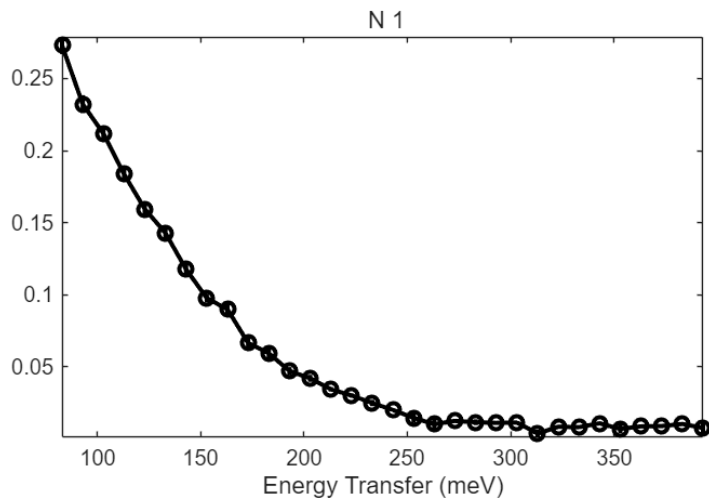
all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);

```

```

bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

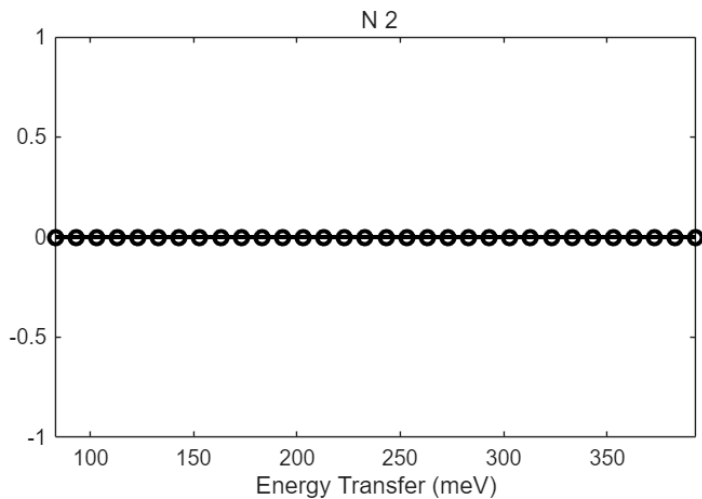
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure

```

```

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);

```

```
fit_src_struct = GH_FFCutsFits;  
save(CutFitFileGH,"fit_src_struct",-v7.3');  
close all;  
%end
```